SGMRS: Spherical Gaussian Guided Multi-resolution Sampling for Fast Foveated 360-degree Video Streaming

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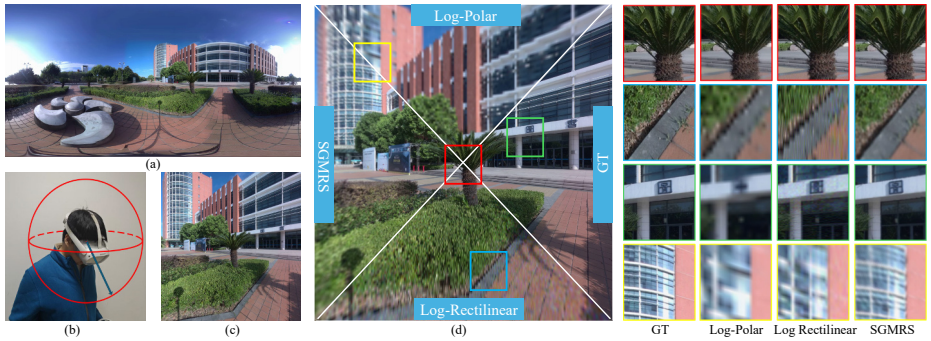


Figure 1: Left: Overview of foveated 360° video streaming. For a frame of the 360° video (a), a multi-resolution foveated viewport frame (c) corresponding to the user's gaze direction (b) is transmitted. Middle (d): Visual comparison between the proposed SGMRS method (left) and the baseline approaches, including Log-Polar (top), Log-Rectilinear (bottom), and the ground truth (GT, full-resolution transmission, right). Right: Close-up views of the reconstructed client-side images. The results show that SGMRS produces images closer to the ground truth, particularly in peripheral regions where our method preserves clearer structural details.

Abstract

With the rapid development of virtual reality technology, 360° panoramic video has become increasingly widespread. To ensure high visual quality across the wide field of view, the resolution of such videos continues to increase, which brings significant challenges for video storage, processing, and especially transmission. Previous research has proposed foveated image transformation techniques to reduce the resolution in non-foveal regions, thereby lowering transmission bandwidth. However, these approaches operate on ERP-projected panoramic video frames without accounting for the severe geometric distortions introduced by ERP projection, particularly in high-latitude regions. This paper proposes a spherical Gaussian-guided multi-resolution sampling (SGMRS) method. Specifically, we use a normalized spherical Gaussian distribution to generate a Fibonacci-based sampling point set that conforms to the spherical visual acuity model, and perform foveated multi-resolution sampling directly in the spherical coordinate domain. We then precompute the spherical sampling map for a canonical gaze direction and exploit the rotational symmetry of the spherical Gaussian distribution to achieve fast recovery of 360° foveated video frames. This supports efficient per-frame processing for gaze-adaptive 360° video streaming We conduct quantitative evaluations on existing 360° video datasets and compare our method against state-of-the-art techniques. The results demonstrate that our method substantially improves visual consistency across different gaze directions. Under the evaluated settings, SGMRS improves foveal-region quality by up to 11.24 dB in FPSNR and reduces flicker by up to 46.7% compared with the baseline methods.

Index Terms: 360° Video streaming, Spherical Gaussian, multi-resolution sampling, foveation, virtual reality

1 Introduction

The advancement of virtual reality (VR) has established 360° video as a key media format, garnering significant interest from users and developers alike. By capturing a full 360°×180° field of view, also known as panoramic video, it offers a highly immersive experience. This immersion, however, comes at a cost: the extensive visual data demands very high resolutions to maintain quality, which in turn poses significant challenges for transmission. Since most pixels in a 360° frame lie outside the user's immediate field of view, conventional 2D video streaming solutions are inadequate for handling the high-resolution video frames produced [7, 30, 29]. Consequently, techniques such as super-resolution [3, 4, 18, 26, 34, 35, 11], viewport-adaptive streaming [10, 25, 24], and multi-resolution scaling [12, 20, 32] are essential to reorganizing and encoding the video stream, thereby reducing the requisite transmission bandwidth.

With advances in eye tracking and multi-resolution imaging, gaze-adaptive video transmission has been widely studied to address bandwidth limitations. These methods preserve high fidelity in the viewer's area of focus while reducing the resolution of peripheral regions, thereby lowering transmission bitrate. Currently, there are two main approaches.

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One approach uses tiling, dividing 360-degree videos into independently streamed tiles encoded at multiple resolutions [33, 20, 9]. However, this requires synchronizing the storage, transmission, decoding, rendering, and blending of multiple streams, imposing substantial burdens on both servers and clients. Another approach uses foveated image transformation, which simulates the human visual system by reducing the resolution of peripheral regions. Originally developed for graphics rendering [8], it has been extended to multi-resolution image processing [23, 41] and 360-degree video transmission [16]. However, directly applying these methods to Equirectangular Projection (ERP) videos can cause severe distortion, particularly when the gaze point lies in high-latitude regions.

In this paper, we propose a spherical Gaussian-guided multi-resolution sampling (SGMRS) method to address the aforementioned problems. Our approach directly models visual acuity on the sphere using a normalized spherical Gaussian distribution. This model guides the construction of a spherical Fibonacci sampling pattern centered on the user's current gaze direction. By precomputing a canonical sampling pattern and exploiting the rotational symmetry of spherical Gaussians, SGMRS enables efficient per-frame sampling and rapid reconstruction under arbitrary gaze motions, while avoiding the need for per-frame re-optimization or complex coordinate transformations. The extensive experiments on 4K and 8K 360° video datasets demonstrate that SGMRS achieves superior perceptual quality in the foveal region while significantly reducing transmission bandwidth, supporting its potential for efficient gaze-adaptive 360° video streaming. Fig. 1 presents a visual comparison of different foveated video transmission methods on the video from the Agtzidis et al. dataset. As observed, SGMRS preserves fine details within the foveal region while simultaneously maintaining higher structural fidelity in the peripheral areas. This advantage is particularly evident in the zoomed-in region on the right side of the rendered image, where SGMRS produces sharper textures and fewer distortion artifacts compared to the other foveated transmission methods. The main contributions of this paper are threefold.

- We introduce a foveated sampling strategy that models human visual acuity directly on the sphere using a normalized spherical Gaussian distribution to guide a Fibonacci sampling pattern. This formulation mitigates the geometric distortion issues that arise in high-latitude regions under ERP projection.
- We propose an efficient gaze-adaptive resampling and reconstruction method that precomputes a canonical sampling pattern and dynamically applies spherical rotations based on the user's current gaze direction.
- We integrate our sampling strategy into a prototype 360° video streaming pipeline and demonstrate, through extensive experiments on 4K and 8K datasets, that our method improves fovea region perceptual quality under the same bandwidth and significantly reduces required bandwidth for equal perceived quality.

2 Related Work

In this section, we briefly review prior work on super-resolution (SR)-based and foveated 360° video streaming. These approaches share the common goal of reducing bandwidth while maintaining perceptual quality. For more in-depth and systematic reviews of this research area, we refer the reader to the survey studies [14, 38].

2.1 SR-based 360° video streaming

Super-resolution (SR)-based 360° video streaming approaches [2, 42] transmit low-resolution (LR) video tiles along with an SR model from the video server to the client. The client then performs SR inference to reconstruct high-resolution (HR) video frames, thereby reducing the bandwidth consumption required for streaming high-quality 360-degree video content.

In recent years, many SR-based video streaming systems have been proposed. Chen et al. [2] reduce the bandwidth consumption of 360° video streaming by training SR models at the tile level and transmitting only low-resolution tiles to the client. Similarly, PARSEC [3] achieves high-quality 360° video streaming by employing lightweight SR models trained on short video segments and jointly optimizing the allocation of network and computational resources to reduce both model size and inference cost. Liu et al. [11] further present an SR-based 360° video conferencing system that incorporates SR models to enhance streaming quality while maintaining low inference latency. More recently, Masked360 [21] introduces a robust neural 360° streaming framework, in which a lightweight masked autoencoder variant (MaskedEncoder) is transmitted to the client. This model extracts latent representations from partially masked low-resolution frames and reconstructs perceptually enhanced full-resolution frames.

Although SR-based 360° video streaming methods can reduce transmission bandwidth to some extent, they still suffer from several limitations. On the one hand, large upscaling factors often lead to noticeable degradation in reconstructed video quality [13]. When smaller upscaling factors are used, the bandwidth reduction becomes limited, especially considering the inherently high bandwidth requirements of 360° video streaming. On the other hand, many SR models involve computationally intensive and time-consuming inference processes [17], making real-time super-resolution challenging and requiring expensive hardware support.

2.2 Foveated 360° video streaming

Another widely adopted class of approaches reduces the bandwidth requirements of high-resolution video streaming through multi-resolution techniques. Early methods typically relied on tiling strategies combined with viewport prediction, where only the visible region of the video is transmitted at high quality, while the non-visible regions are delivered at lower quality or even entirely omitted [20, 7, 10, 28, 24]. For example, Qian et al. developed Flare [29], which applies tiling to viewport-adaptive streaming with the goal of delivering the entire viewport of a 360° video at full resolution over mobile networks. Ozcinar et al. [27] designed a 360° video streaming system that incorporates a viewport-aware bitrate selection algorithm. Yaqoob et al. [40] proposed a tile-based adaptive streaming solution called CFOV, which integrates two field-of-view prediction mechanisms and a priority-based bitrate adaptation strategy to ensure effective bandwidth allocation. Wu et al. [39] introduced a 360° video streaming framework that uses deep reinforcement learning to determine the optimal download order and bitrate for tiles based on user preference. Although tile-based techniques can effectively eliminate the transmission of out-of-viewport regions, they face practical limitations when high-quality variable-resolution rendering is required. If the number of tiles is small, tearing and color discontinuity artifacts may appear at tile boundaries. However, increasing the number of tiles substantially raises the burden on client devices, which must simultaneously stream, decode, and render hundreds of video segments, while also performing blending operations to suppress boundary artifacts.

Later, with the advancement of foveated rendering techniques [8, 23], a series of foveated multi-resolution 360° video encoding methods emerged. For example, Li et al. [16] proposed a log-rectilinear transformation that preserves full-resolution fidelity in the foveal region while smoothly reducing the resolution toward the periphery. By integrating the summed-area table, gaze positions, and off-the-shelf video codecs, their system enables foveated 360° video streaming for eye-tracking VR head-mounted displays. However, when these methods are applied to ERP-projected 360° videos, they do not account for the geometric distortions introduced

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at high latitudes. As a result, the transformed images suffer from significant quality degradation, which is particularly pronounced for high-resolution 360° videos.

3 Analysis

As previously indicated, applying foveated image transformation techniques [23, 16] to 360° videos results in significant distortion in the transformed image when the user's gaze is directed towards high-latitude regions. In panoramic imaging, spherical panoramic images are typically stored and transmitted by being mapped to a rectangular format using the Equirectangular Projection (ERP) [6]. This projection maps meridians to vertical lines with constant spacing and parallels of latitude to horizontal lines with constant spacing. However, this projection introduces distortion, which becomes increasingly pronounced in regions closer to the poles.

Prior research has achieved 360° foveated video streaming by leveraging log-polar and log-rectilinear transformation techniques. These methods detect the user's gaze coordinates and partition the panoramic image into multiple sampling regions. Centered around the user's gaze point, these regions are sampled at progressively lower rates with increasing distance from the focal point. The resulting transformed image prepared for transmission reduces frame information in non-foveal areas while preserving pixel samples in the foveated region, thereby enabling efficient foveated video streaming.

However, these methods suffer from a significant limitation. When the user's gaze point is located near the equatorial region of the panoramic video frame—where distortion in the ERP projection is minimal—the delineated foveated region in both methods remains closely aligned with the actual area of visual focus. Conversely, when the user's gaze point approaches the polar regions of the frame, where geometric distortion becomes severe, the foveated region generated by these methods deviates substantially from the actual area of interest, as illustrated in Fig. 2. Fig. 2 top shows the

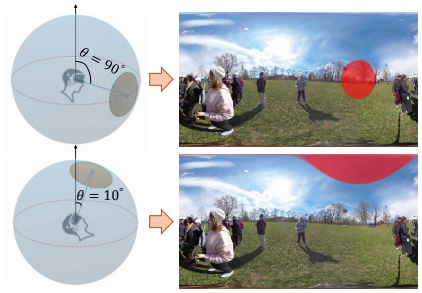


Figure 2: Illustration of the foveal region projection under different gaze directions. When the user's gaze is directed near the equator (top, $\theta = 90°$) or close to the poles (bottom, $\theta = 10°$), the corresponding spherical foveal region maps to the red-highlighted areas on the ERP image (right). Due to the non-uniform distortion inherent in the ERP projection, the projected foveal regions vary significantly in shape and spatial extent across different gaze latitudes.

actual foveated region (marked with red) in the panoramic image when the user's gaze is near the equator in a level viewing state. Fig. 2 bottom shows the actual foveated region when the user looks upward towards a polar region (with a polar angle $\theta = 10°$). It can be observed that, due to the distortion inherent in the panoramic image, a significant deviation occurs in the image sampling of the foveated region when the user's gaze is directed at high-latitude areas. Since both the log-polar and log-rectilinear methods implicitly assume that the foveal region corresponds to a compact area, as illustrated by the red region in Fig. 2 top, they fail to account for the distortion that occurs when the gaze point approaches the poles. As a result, when the user's gaze direction shifts toward high-latitude regions, the image quality within the foveal region degrades significantly after transmission.

We conducted comparative experiments on the panoramic image (4K, 3840×1920) in Fig. 3, applying log-polar, log-rectilinear transformations (reduced-resolution buffer size: 1072×608) and our SGMRS sampling methods (sampling num: 500k) at user gaze points corresponding to polar angles of $\theta = 90°$ and 180°. The resulting foveated multi-resolution images are shown in Fig. 3. It can be observed that when the user's gaze shifts from the equatorial region (top row) to the polar region (bottom row), the foveal region (highlighted by the red circle) in the reconstructed images produced by the log-polar and log-rectilinear transformations suffers a noticeable quality degradation. In contrast, our SGMRS method preserves the visual quality of the foveal region effectively and even achieves slight improvements. Moreover, despite using the fewest sampling points among all methods, our approach consistently provides significantly better reconstructed image quality from the user's viewpoint across different gaze directions.

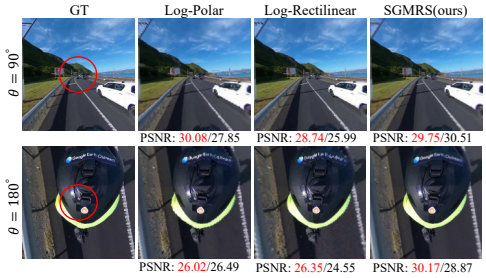


Figure 3: Comparison of user-view reconstructed image quality under equatorial and polar latitude gaze conditions. Three methods (log-polar, log-rectilinear, and our SGMRS) are compared with the ground-truth (GT) frames, and the PSNR values for fovea region (within the red circle masked) and entire image are reported.

Based on the aforementioned analysis and validation, we have identified that existing foveated 360° video transmission methods fail to account for the distortion introduced by the ERP projection in panoramic video frames. This oversight leads to noticeable visual artifacts when the user's gaze point approaches high-latitude regions. Therefore, it is imperative to develop a novel foveated multi-resolution transformation and transmission method capable of consistently maintaining high-quality foveated regions across all orientations in 360° videos.

4 Method

4.1 Framework Overview

Unlike directly sampling from ERP-mapped panoramic images, we aim to maintain a consistent visual presentation when the user explores the scene in all directions. To this end, we propose a spherical Gaussian guided multi-resolution sampling (SGMRS) method, which performs foveated multi-resolution sampling directly on the sphere. Our SGMRS method models the user's spherical visual acuity using a spherical Gaussian distribution, which guides multi-resolution sampling of 360° videos on the server side. By constructing a precomputed sampling distribution map, our method enables fast reconstruction of 360° video on the client side, thereby supporting efficient processing for foveated 360° video streaming.

Our work builds upon the foveated video streaming pipeline proposed by Li et al.[16]. The key difference is that, in our system, we incorporate the spherical visual acuity model and perform foveated multi-resolution sampling directly on the decoded panoramic video

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frames at the server side. On the client side, the transmitted sampled video frame is efficiently reconstructed by applying the precomputed canonical spherical sampling pattern and adjusting it according to the current fovea direction. The pipeline of SGMRS is illustrated in Fig. 4 and introduced below.

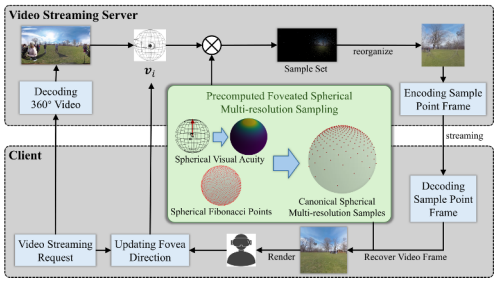


Figure 4: The system pipeline of our SGMRS video streaming prototype. We extend the pipeline of Li et al. [16] by incorporating an additional precomputation module, which serves both the server and the client for fast sampling and frame recovery.

As illustrated in Fig. 4, our sampling method relies on a precomputed foveated spherical multi-resolution sampling module. By combining this module with the user's actual gaze direction $\mathbf{v}_i$, we achieve fast 360-degree video frame sampling on the server side and efficient viewpoint image reconstruction on the client side, thereby enabling foveated 360° video streaming. In Sec. 4.2 and Sec. 4.3, we provide a detailed description of the proposed foveated spherical multi-resolution sampling method, along with its corresponding implementation details, including canonical sampling, frame reorganizing and recovery.

4.2 Foveated Spherical Multi-resolution Sampling

4.2.1 Spherical Visual Acuity Model

In the human visual system, visual acuity is not uniformly distributed across the entire field of view. The foveal region of the eye has much higher visual acuity, allowing fine details in the foveated viewing area to be recognized when a person fixates on an object, whereas peripheral visual regions cannot be perceived with the same level of clarity [36]. In the context of 360° video streaming, we adopt a spherical Gaussian distribution to construct our spherical visual acuity model for the following reasons.

- First, the integral of the spherical Gaussian distribution has a closed-form solution. Consider the following spherical Gaussian distribution function:

$$G(\mathbf{v};\mu,\lambda,a) = ae^{\lambda(\mu\cdot\mathbf{v}-1)} \tag{1}$$

  We can compute the integral Z of this spherical Gaussian distribution function over the entire sphere, obtaining:

$$Z = \int_{\Omega} G(\mathbf{v};\mu,\lambda,a) = 2\pi\frac{a}{\lambda}(1-e^{-2\lambda}) \tag{2}$$

  here, μ is the axis or direction of the lobe. It effectively controls the position of the lobe on the sphere and always points to the center of the lobe. λ represents the sharpness of the lobe. As this value increases, the lobe becomes *narrower*, meaning that the function value decreases more rapidly as one moves away from the lobe axis. a denotes the amplitude or intensity of the lobe. Given that the spherical Gaussian distribution admits a closed-form solution for its integral, we can easily normalize it and compute spherical sampling rates through inverse cumulative distribution function (CDF) mapping. This allows us to determine the sampling distribution for 360-degree video content.

- Second, the spherical Gaussian distribution exhibits rotational symmetry on the sphere, and rotating a spherical Gaussian is straightforward—one only needs to apply a rotation to the axis vector of the spherical Gaussian. The rotation can be performed using matrices, quaternions, rotation vectors, or any other standard representation. This property enables the sampling point set to be efficiently updated during sampling by applying a rotation transformation based on changes in the user's gaze direction.

4.2.2 Gaussian guided Spherical Fibonacci Points

To reflect the perceptual characteristics of spherical visual acuity, we directly generate sampling points on the sphere based on a spherical Gaussian distribution centered on the foveated direction. We first construct an approximately uniform point set $\{v_i\}$ on the unit sphere $\mathbb{S}^2$ using spherical Fibonacci sampling. The spherical Fibonacci lattice provides quasi-uniform coverage with minimal sampling artifacts and avoids the distortions and pole singularities inherent to ERP-based parameterizations. This ensures rotational invariance of the sampling grid, which is crucial for maintaining stable rendering quality under head rotations in VR.

Next, we guide the sampling density using our spherical Gaussian visual acuity model. For each point v_i, we compute its sampling weight as:

$$w_i = P(v_i|\mu,\lambda) = \frac{G(\mathbf{v};\mu,\lambda,a)}{Z} \tag{3}$$

Higher values of λ result in sharper fovea emphasis, while lower values yield smoother transitions between high and low resolution regions.

We then perform importance sampling over $\{v_i\}$ using the weights $\{w_i\}$ to obtain a final sampling point set:

$$V_{\text{SGMRS}} = \text{SampleFibonacci}(\{v_i\},\{w_i\}) \tag{4}$$

This results in a multi-resolution sampling distribution that naturally concentrates sampling density near the foveated direction while reducing sampling rate in peripheral regions.

To determine the appropriate setting of the parameter λ in the sampling process, we conducted pre-sampling experiments on 2K, 4K, and 8K 360° video frames. Specifically, we evaluated the client video quality under different numbers of sampling points N and different values of λ. The goal of this evaluation is to identify optimal sampling parameters that minimize the number of sampling points while preserving the user's perceived visual quality. Fig. 5 shows one frame of panoramic sampling results and the reconstructed client viewport images for the 2K, 4K, and 8K video. The PSNR/FPSNR values of the recovered image (2048×2048) are reported. All pre-sampling experimental data are shown in Tab. 2. In practice, we choose the parameters setting as $N = 500k$, $\lambda = 5.0$.

4.3 Implementation details and Precomputation

After performing foveated spherical multi-resolution sampling, a set of non-uniformly distributed sampling points is obtained in the spherical coordinate domain. Direct transmission of these samples, however, is inefficient for two reasons. First, transmitting both pixel values and their associated panoramic coordinates introduces substantial bandwidth overhead. Second, the irregular spatial distribution of the samples disrupts spatial coherence, significantly degrading video compression efficiency and thereby increasing the required transmission bitrate. To overcome these limitations, we design a transmission strategy that leverages the structural properties of the normalized spherical Gaussian distribution.

First, we exploit the rotational symmetry of spherical Gaussians to eliminate the need for transmitting sampling coordinates. A canonical sampling pattern is precomputed with respect to a fixed

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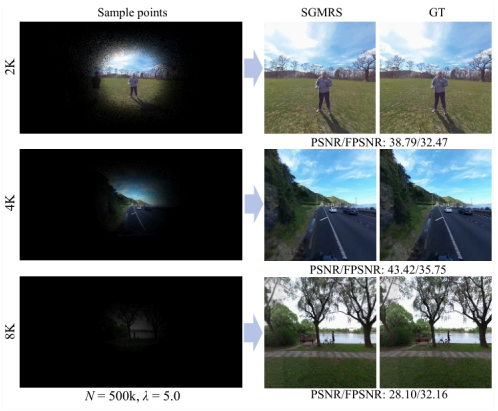


Figure 5: Visualization of sampling points, reconstructed user-view frame, and ground-truth user-view frame for 360° videos at different resolutions. The PSNR/FPSNR values are reported.

canonical gaze direction. For each frame, the sampling positions corresponding to the current gaze direction are obtained via spherical rotation using directional offsets. Consequently, only sampled pixel values are transmitted, while sampling coordinates are implicitly reconstructed at the receiver, effectively reducing transmission overhead.

Second, to improve compression efficiency, we reorganize the sampled pixels into a structured square frame centered at the gaze direction prior to encoding. Since spherical Gaussian-guided sampling produces a dense distribution around the foveal region, this reparameterization preserves local spatial continuity among neighboring samples to a large extent. The resulting structured representation is significantly more amenable to conventional video codecs, leading to improved compression efficiency and reduced bandwidth consumption without compromising foveal perceptual quality.

4.3.1 Canonical Sampling

We precompute a canonical sampling point set on the unit sphere for a reference gaze direction $\mathbf{v}_0$. Let the precomputed set be

$$\mathscr{S}_0 = \{\mathbf{s}_j \in \mathbb{S}^2 | j = 1, 2, ..., N\}, \tag{5}$$

where each $\mathbf{s}_j$ is a unit direction vector expressed in the spherical (Cartesian) coordinate system and $\mathbf{v}_0 \in \mathbb{S}^2$ denotes the canonical gaze axis (for example, the north pole). At runtime, when the user's gaze direction is $\mathbf{v}_i \in \mathbb{S}^2$, the actual sampling points for this gaze are obtained by rotating the precomputed set $\mathscr{S}_0$ so that the canonical axis $\mathbf{v}_0$ is mapped to $\mathbf{v}_i$. In practice, we use a quaternion $\mathbf{k}$ representation to apply the rotation for stability and compactness, which can be calculated as:

$$\begin{cases} \theta = \arccos(\cos(\mathbf{v}_0 \cdot \mathbf{v}_i)) \\ \mathbf{k} = \frac{\mathbf{v}_0 \times \mathbf{v}_i}{||\mathbf{v}_0 \times \mathbf{v}_i||} \\ \mathbf{q} = (\cos(\frac{\theta}{2}), \mathbf{k}\sin(\frac{\theta}{2})) \end{cases} \tag{6}$$

Here, θ is rotation angle, and $\mathbf{k}$ is the unit rotation axis between $\mathbf{v}_0$ and $\mathbf{v}_i$. Applying it to every precomputed sample, we can recover the final image frame by getting the rotated sampling points $\mathscr{S}_i$ as

$$\mathscr{S}_i = \{\mathbf{s}'_j = \mathbf{q}\mathbf{s}_j\mathbf{q}^{-1} | \mathbf{s}_j \in \mathbb{S}^2\}. \tag{7}$$

4.3.2 Reorganizing Streaming Frame

The resulting sampling set $\mathscr{S}_i$ is a discrete collection of pixel points on the sphere. To integrate our method into the video streaming pipeline of Li et al. [16], we reorganize the sampled point set into a square frame centered at the current gaze direction. In the resulting frame, each pixel corresponds to one sampled point.The reorganized frame is fixed to be square, and its side length is selected as the smallest integer resolution that provides enough pixels to accommodate all sampling points. Specifically, for N sampling points, we set the side length to $L = \lceil N \rceil$, so that the reorganized frame contains at least N pixel locations.

As illustrated in Fig. 6, the square frame $\mathbf{I}$ is defined on a spherical tangent plane P centered at the user's foveated point o and perpendicular to the gaze direction $\mathbf{d}$, forming a 90° field of view frustum. This construction establishes a local image plane aligned with the gaze direction, enabling a structured 2D parameterization of the spherical samples. Subsequently, the sampled points are rearranged

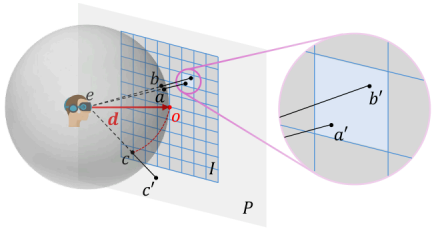


Figure 6: Illustration of the frame reorganization process. All spherical sampling points are rearranged onto the frame image shown in $\mathbf{I}$ according to Alg. 1, while preserving the structural coherence of the original sampling pattern as much as possible.

within the frame image according to Alg. 1, which deterministically maps the spherical sampling pattern onto the square grid. This reorganization preserves spatial coherence among neighboring samples while maintaining consistency with the precomputed canonical sampling pattern, thereby facilitating efficient encoding within the existing streaming framework.

Algorithm 1: Sample Points Reorganizing

Input: Sample point set $\mathbf{S}$, User view direction $\mathbf{d}$, User eye pos e
Output: Reorganized sample point frame $\mathbf{I}$

```
1  QuickSort(S, d);
2  for i = 1 to S.Size() do
3      (x,y) = Projection(S_i, d, I);
4      if x < I.col & y < I.row & IsEmpty(I(x,y)) then
5          I(x,y) = S_i.color;
6          S = S − {S_i};
7      end
8  end
9  for j = 1 to S.Size() do
10     (x,y) = Projection(S_j, d, I);
11     (x',y') = FindClosestEmptyPixel-KNN(I, e, S_j, (x,y));
12     I(x',y') = S_j.color;
13 end
14 return I
```

Alg. 1 consists of two sequential passes. First, all sampling points are sorted in ascending order according to their spherical distance to the gaze direction ,prioritizing foveal samples to better preserve local spatial structure. (line 1). Then, In the first pass (lines 2-8), we traverse the sorted samples and project each sampling point onto the square frame defined on the tangent plane (line 3), as illustrated in Fig. 6, by intersecting the ray from the sphere

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center through the sampling point with the frame plane. If the corresponding pixel location in the frame is unoccupied, it is assigned as the reorganized position of the sampling point (lines 4-7), like the point a and pixel a' in Fig. 6. If the projected pixel is already occupied (b' in Fig. 6), or if the projection falls outside the valid frame region (c' in Fig. 6), the sample is temporarily deferred to the second pass. In the second pass (lines 9-13), we process the remaining unassigned sampling points. For each such sample, we perform a KNN search in pixel space to locate the closest unoccupied pixel in terms of spherical distance (lines 10-11). This pixel is then assigned as the reorganized position of the sampling point (line 12). After the two passes, all sampling points are deterministically mapped to unique pixel locations in the transmission frame (line 14). if there are still empty pixels in the reorganized frame, these pixels will be assigned the values of their nearest sampling points to facilitate video encoding and compression. These auxiliary pixels are not used in the actual reconstruction.

The first pass preserves the intrinsic spatial structure of the spherical sampling pattern to the greatest extent possible by prioritizing geometrically consistent projections. The second pass guarantees completeness of the mapping while minimizing structural distortion introduced during reorganization. Importantly, the entire reorganization procedure is executed during a precomputation stage. At runtime, the precomputed mapping is directly loaded, introducing no additional computational overhead to the streaming process.

4.3.3 Client Recovery

To generate a continuous video frame from the user's viewpoint, image recovery is required. For unsampled pixels within the viewport, we compute their final color values through interpolation using their four nearest sampled pixels. Since our spherical Fibonacci sampling points remain fixed relative to the gaze direction, their projected coordinates in the user's viewpoint frame are also invariant. To enable fast reconstruction of user-view video frames, we precompute the interpolation parameters corresponding to each pixel in the user-view frame. During runtime, the interpolation results can then be efficiently obtained by directly reading these precomputed parameters, enabling fast reconstruction of user-view frames.

5 Evaluation

We evaluate the potential advantages and limitations of our method through a quantitative assessment of the visual quality of the user's viewpoint video and bandwidth savings. We compare our approach with log-rectilinear transformation with summed-area table sampling [16] and the commonly used log-polar transformation in foveated rendering [23]. Additionally, we conduct ablation experiments on several important parameters of our method. In the supplementary video, we demonstrate a comparison of the final user viewpoint video effects between our method and the two aforementioned comparison methods.

5.1 Dataset and Configuration

To validate our method and ensure reproducibility, we conducted a quantitative evaluation of our video streaming system using the benchmark 360° video dataset by Agtzidis et al. [1], following the Log-rectilinear method [16]. This dataset contains 14 different 360° videos, with gaze points and head trajectories recorded from 13 participants using the FOVE3 eye-tracking VR headset. We performed a comparison of quality and bandwidth for all 172 (participant, video) pairs, and reported the aggregated average metrics. Since the highest video resolution in the Agtzidis's dataset is 4K (3840×2160), we also evaluated our video streaming system using the 8K video dataset from Liu et al [19]. This dataset includes 12 different 360° videos. We transferred the 12 gaze points and head trajectories from the Agtzidis dataset to this new dataset and conducted a comparison of quality and bandwidth for all 144 (participant, video) pairs, and reported the aggregated average metrics.

All experiments were conducted on both server and client systems equipped with an Intel Core i9-13900K CPU and an NVIDIA RTX 4090 GPU. In our benchmarking, decoding was performed on the CPU using x264, while encoding was done on the GPU using NVIDIA NVENC and the FFmpeg API. All experiments and measurements were conducted on panoramic video ERP projection frames at 2K (2560×1440), 4K (3840×1920), and 8K (8192×4096) resolutions. For log-Polar and SAT log-rectilinear methods, we used a 1072×608 resolution buffer. For our method, we sampled 500k points for 2K, 4K and 8K videos. The 2K and 4K videos were from the Agtzidis et al. dataset [1], with the 4K videos being raw panoramic footage and the 2K videos being pre-processed and downscaled (consistent with the Log-Rectilinear method [16] for easier comparison). The 8K videos were sourced from the Liu et al. dataset [19]. During the sampling phase, for the Log-Polar method, we applied a 3×3 Gaussian blur to the right half of the logarithmic polar encoding buffer following Meng et al.'s approach [23]. Our method and the Log-Rectilinear method did not apply any blur.

5.2 Quality

Fig. 7 presents a visual comparison of different foveated video transmission methods on four randomly selected 360° videos (two 4K and two 8K) from the two datasets. The resolution of the reconstructed client-side images is 2048 × 2048. The red dashed circle in the images indicates the foveal region, while the red, yellow, blue, and green squares correspond to the four zoomed-in regions. Overall, our method reconstructs visual results in the foveal region that are close to the ground truth (GT). At the same time, compared with the baseline methods, SGMRS preserves peripheral details more effectively.

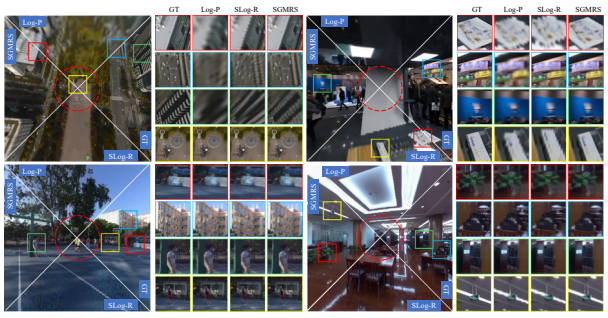


Figure 7: Visual quality comparison results. The top row correspond to videos with a 4K resolution, while the bottom row correspond to 8K videos. For each scene: Left: Comparison between the proposed SGMRS method (left), the baseline approaches—including Log-Polar (Log-P, top) and SAT Log-Rectilinear (SLog-R, bottom)—and the ground truth (GT, full-resolution transmission, right). Right: Zoomed-in views of the reconstructed images. The results show that SGMRS produces images closer to the ground truth and generally preserves clearer structural details, especially for periphery regions.

For the top row (4K scenes), the GT images themselves exhibit slight blur due to the limited source resolution. The baseline methods tend to amplify this blur, whereas our method produces results that are closer to the GT in the central region. In peripheral regions, although some additional blur can still be observed, SGMRS better preserves structural information in the scene. For example, vehicles on the road remain clearly distinguishable (second row in the city scene), and the edges of the booklet appear sharper (first row in the indoor scene). In contrast, the baseline methods significantly increase blur and introduce noticeable aliasing artifacts.

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For the bottom row (8K scenes), artifacts become more evident in the peripheral regions for the baseline methods. The Log-Polar approach produces pronounced blur artifacts, while SAT Log-Rectilinear introduces horizontally distributed texture artifacts. These issues are likely caused by excessive compression of peripheral regions during the transformation-based sampling process. In comparison, SGMRS maintains noticeably better detail preservation in the reconstructed images.

To assess the visual quality of our method, we compared the differences between the foveated video and the original video in the final user viewpoint under constant quality mode. For global quality assessment, we adopt Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) to measure the average quality over the entire frame. To specifically evaluate the foveation quality, we employ Foveated Peak Signal-to-Noise Ratio (FPSNR) [15, 31]. In our evaluation, FPSNR is computed with a gaze-centered Gaussian weighting function, where the weight of each pixel is defined as $w = \exp(-(e^2)/(2\sigma^2))$. Here, e denotes the angular distance between the pixel direction and the gaze direction in degrees, and σ is set to 2° in all experiments. We further employ FovVideoVDP [22] to account for the spatially varying characteristics of human visual sensitivity. These metrics jointly provide both objective full-frame fidelity evaluation and perceptually grounded foveated quality assessment.

Tab. 1, columns 3-6 reports the FPSNR, PSNR, SSIM, and FovVideoVDP results of different foveated video streaming methods. Compared with log-polar and log-rectilinear transformations, SGMRS achieves the best or highly competitive performance across the evaluated metrics in most cases, demonstrating its effectiveness in preserving perceptual quality under foveated streaming.

Table 1: Visual quality and temporal perception quality comparisons of Log-Polar, SAT Log-Rectilinear, and our SGMRS method based on the aggregate average values of FPSNR, FVVDP, PSNR, SSIM and Flicker metrics on Agtzidis et al. dataset (for 2K and 4K) and Liu et al. dataset (for 8K). SGMRS (proposed) generally provides the best video streaming quality (marked bold).

Res.	Method \ Metric	FPSNR(db)↑	FVVDP↑	PSNR(db)↑	SSIM↑	Flicker↓
2K	Log-P	29.04	6.28	30.15	0.92	449.34
	SLog-R	27.82	6.19	28.82	0.90	497.45
	SGMRS	**38.41**	**7.39**	**33.57**	**0.95**	**339.31**
4K	Log-P	30.99	5.91	28.80	0.91	506.30
	SLog-R	29.36	5.92	28.07	0.88	605.92
	SGMRS	**42.23**	**8.13**	**35.43**	**0.96**	**275.65**
8K	Log-Polar	34.78	5.35	27.93	0.87	462.51
	SLog-R	32.37	5.40	28.65	0.80	620.18
	SGMRS	**39.09**	**8.30**	**35.67**	**0.92**	**246.57**

Following the setup by Li et al. [16], we also adopted the metrics proposed by Winkler et al. [37] to quantify temporal instability. The flicker value is computed for all consecutive frame pairs in a video, and the final flicker score is obtained by averaging these values over the entire sequence. Since each video in the dataset contains a single scene, all frames are assigned equal weights when computing the overall flicker score. Tab. 1, Column 7 reports the Flicker metric for different foveated video transmission methods. The proposed SGMRS method demonstrates improved temporal consistency across the entire dataset, effectively reducing video flicker introduced during the transmission process. Compared with Log-P and SLog-R, SGMRS achieves the lowest flicker values.

5.3 Ablation

In our SGMRS method, two key parameters influence the quality of the transmitted and reconstructed user-view video: the number of sampling points N and the parameter λ in the spherical Gaussian distribution function (Eq. (1)). A larger N results in more sampled pixels per frame, leading to higher image quality but increased transmission load. Conversely, a larger λ makes the spherical Gaussian probability distribution more concentrated around the fovea center, improving the quality of the fovea region in the reconstructed frame, while the image quality of the peripheral regions may degrade.

In this section, we conduct two ablation experiments to evaluate the effects of the two parameters, N and λ, in SGMRS on the reconstructed user-view video quality under different panorama resolutions. (1) Fixing $\lambda = 5.0$, we assess the recovered frame quality under different sampling point counts: $N = 10k, 50k, 100k, 300k, 500k$, as shown in Fig. 8, (a). (2) Fixing the sampling point count $N = 500k$, we evaluate the recovered frame quality under different values of $\lambda = 1.0, 2.5, 5.0, 7.5, 10.0$, as illustrated in Fig. 8, (b). Additionally, Tab. 2 summarizes the reconstructed image quality under all parameter configurations. The results show that the recovered image quality consistently improves as the number of sampling points increases. As the parameter λ increases, the reconstruction quality exhibits different trends across video resolutions. For the 2K videos, both PSNR and FPSNR reach their maximum values at λ = 5.0, indicating the best overall reconstruction quality under this setting. For the 4K and 8K videos, the PSNR metric also peaks at λ = 5.0. However, the FPSNR metric continues to increase as λ grows and reaches its maximum at λ = 10.0. Considering both bandwidth constraints and ablation results, we select the parameter configuration highlighted in bold ($\lambda = 5.0$, $N = 500k$) as the optimal setting for quality and bandwidth evaluation.

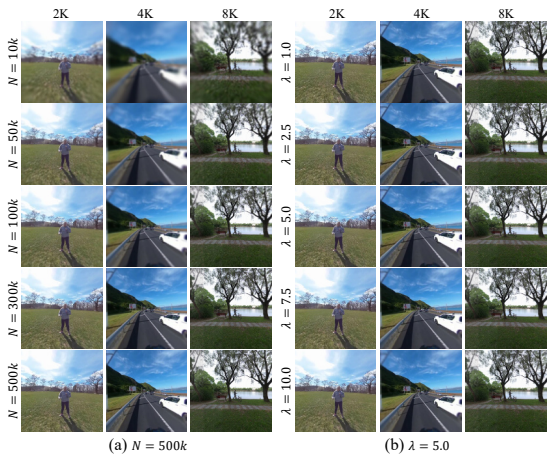


Figure 8: Ablation results shown as reconstructed 2D user viewport images with a 90°×90° FOV: (a) N values when $\lambda = 5.0$, and (b) λ values when $N = 500k$.

5.4 Bandwidth

We measured the average bitrate of the H.264 stream generated by FFmpeg to determine the compression ratio achieved by our proposed SGMRS method. All processes were encoded using the H.264 main profile, with the constant quality parameter (cq) set to 25. Tab. 3 shows the bitrate for each process and the corresponding per-frame packet size.

As shown in Table 3, SGMRS exhibits transmission performance comparable to that of the baseline methods. This outcome is closely related to the sampling configuration adopted in our experiments. Specifically, the number of sampling points per frame in SGMRS (500k) is comparable to the intermediate transmitted frame resolution used by the comparison methods (1072 × 608). Moreover,

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Table 2: Quantitative comparison of reconstructed user-view video under different 360° video resolutions (2K, 4K, and 8K) and sampling parameters, including the number of sampling points N and the spherical Gaussian distribution parameter λ. The table reports the PSNR/FPSNR values of the reconstructed video. Bold values indicate the parameter configurations adopted for the quality and bandwidth evaluations. Underlined values correspond to the results of the recovered images used in the ablation studies shown in Fig. 8.

Res.	λ \ N	10k	50k	100k	300k	500k
2K	1.0	27.76/23.10	30.84/26.01	32.74/28.33	36.05/31.76	37.10/32.80
	2.5	28.59/24.14	32.36/28.69	34.35/30.67	37.08/33.16	37.89/33.66
	5.0	29.21/25.34	33.08/30.65	34.99/32.57	37.35/33.75	**38.01/33.67**
	7.5	29.26/26.98	32.82/32.21	34.63/33.06	37.08/33.88	37.73/33.52
	10.0	28.83/27.67	32.41/32.49	34.11/33.54	36.42/33.66	37.27/33.36
4K	1.0	24.17/26.72	27.88/29.19	30.33/31.06	34.86/34.62	36.82/36.28
	2.5	24.99/27.52	29.25/31.29	32.06/33.16	36.45/37.26	38.24/38.45
	5.0	25.19/28.52	29.48/33.18	32.22/36.03	36.50/38.66	**38.25/39.11**
	7.5	24.84/29.76	28.70/35.04	31.11/37.36	35.34/39.18	37.20/39.56
	10.0	24.28/30.20	27.67/35.93	29.70/37.97	33.63/39.66	35.54/40.06
8K	1.0	21.85/29.22	23.65/30.91	24.56/31.98	26.50/34.17	27.85/35.62
	2.5	22.35/29.47	24.12/31.98	25.23/33.45	27.56/36.56	29.12/38.43
	5.0	22.46/30.52	24.16/33.39	25.26/35.19	27.76/39.14	**29.26/41.25**
	7.5	22.27/31.13	23.92/34.40	24.94/36.42	27.10/40.78	28.47/42.49
	10.0	21.73/31.51	23.41/35.15	24.37/37.47	26.21/41.81	27.48/43.14

Table 3: Bandwidth evaluation of different video streaming methods based on the average packet size (kb) and bit rate (Mbps).

Res.	Metric \ Method	Full Res.	Log-P	SLog-R	SGMRS
2K	Aver. Pkt-Size↓	152.98	25.31	28.82	27.98
	Bitrate↓	37.56	6.22	7.08	6.88
4K	Aver. Pkt-Size↓	250.22	24.33	28.80	27.23
	Bitrate↓	61.43	5.98	7.08	6.69
8K	Aver. Pkt-Size↓	874.86	32.74	26.36	17.34
	Bitrate↓	215.00	8.05	6.48	4.26

our transmission frame reorganization algorithm preserves the intrinsic structural coherence of the spherical sampling pattern. By maintaining spatial continuity among neighboring samples, the proposed method mitigates the adverse impact of structural information loss on compression efficiency, thereby preventing unnecessary increases in packet size during transmission. This result indicates that the quality gains achieved by SGMRS do not come at the cost of increased transmission overhead.

5.5 Computational Overhead

To evaluate the computational overhead of our method, we compare the runtime performance of SGMRS with Log-Polar and SAT Log-Rectilinear on both the server side and the client side of the foveated streaming pipeline. As reported in Tab. 4, the server-side processing time of the three methods is divided into four stages: Decoding, Processing, Sampling, and Encoding. As the resolution of the original video increases, the decoding stage becomes increasingly time-consuming for all methods. In the Processing stage, both Log-Polar and SGMRS mainly perform color space conversion on the decoded frames, whereas SAT Log-Rectilinear additionally computes summed-area tables (SAT), which leads to a higher computational cost. In the Sampling stage, SAT Log-Rectilinear and SGMRS perform sampling operations, while Log-Polar mainly applies the log-polar transformation to the video frames, resulting in a relatively larger runtime overhead.

On the client side, we measure the Recover time required to reconstruct the transmitted video into the user's viewpoint. The reconstruction time of SGMRS remains nearly constant across resolutions, while the runtime of Log-Polar and SAT Log-Rectilinear increases noticeably. This behavior is mainly due to the fact that SGMRS reconstructs only the user's visible region rather than the entire panoramic frame. Considering the total runtime, SGMRS consistently achieves the lowest computational cost for 2K, 4K, and 8K videos.

Table 4: Performance comparison (measured in millisecond) of video streaming based on various foveation methods. Compared with the other pipelines, SGMRS consistently exhibits the lowest computational cost across all evaluated resolutions.

Res.	Method	Server				Client		Total
		Decoding	Processing	Sampling	Encoding	Decoding	Recover	
2K	Log-P	3.51	0.81	7.15	5.50	0.54	9.90	27.41
	SLog-R	2.95	8.52	1.34	4.36	0.45	8.46	26.08
	SGMRS	2.89	0.69	1.93	2.34	0.60	7.05	**15.5**
4K	Log-P	10.08	2.93	22.94	4.94	0.59	32.16	73.64
	SLog-R	10.26	35.30	1.37	6.53	0.67	27.67	81.8
	SGMRS	9.88	3.69	10.13	5.62	1.38	8.77	**39.47**
8K	Log-P	60.29	29.35	102.17	3.47	0.33	159.19	354.8
	SLog-R	58.61	166.58	1.37	5.82	1.55	123.78	357.71
	SGMRS	57.19	33.26	17.09	2.70	0.42	8.84	**119.5**

In addition to the online per-frame computation, SGMRS requires a one-time offline precomputation of the canonical sampling and interpolation maps. In our implementation, this process takes approximately 3–5 minutes and is performed only once before streaming. The resulting maps are reused during runtime and therefore do not contribute to the per-frame latency reported in Tab. 4. Their total memory footprint is 142 MB (6MB for the sampling map and 136MB for the interpolation map), representing a fixed storage overhead independent of the number of video frames.

5.6 Discussion

Our experimental results demonstrate that, compared with the log-polar and log-rectilinear methods that perform sampling directly on ERP frames, our SGMRS approach—sampling directly on the sphere—significantly improves the visual quality of the reconstructed video, particularly for high-resolution cases (4K and 8K). This improvement primarily arises from the ability of our method to concentrate sampling points toward the fovea region through the spherical Gaussian distribution parameter λ, ensuring that pixel samples are densely distributed within the user's viewport and thus enabling higher-quality viewport frame reconstruction.

In terms of bandwidth, our method achieves comparable average packet size and bitrate to the baseline methods at 2K and 4K resolutions. However, at 8K resolution, our approach substantially reduces bandwidth consumption. Given this significant reduction in bitrate, we believe that SGMRS has the potential to support future high-resolution gaze-adaptive 360° video streaming systems.

From the ablation studies, we observe that our method can effectively balance the image quality between the fovea region and the entire viewport by adjusting the λ parameter. Moreover, the reconstructed video quality of our method shows weak dependence on the original video resolution, but a strong correlation with the number of sampling points (N). As shown in Tab. 2, this indicates that for high-resolution videos, our approach can maintain stable viewport reconstruction quality without a significant increase in the number of sampling points. Therefore, our method is well-suited to the evolving trend toward higher-resolution 360° video content, offering scalability in both quality and efficiency for future immersive streaming applications.

6 User Study

We conducted a controlled within-subject user study to evaluate the perceptual visual quality of SGMRS on the two aforementioned datasets (4K and 8K), and to compare its performance with existing foveated video streaming methods.

6.1 User Study Design

Participants and Setup. A total of 24 participants (12 male and 12 female, aged between 21 and 30 years) were recruited for the study. All participants had prior experience using virtual reality

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head-mounted displays (VR HMDs)and had normal or corrected-to-normal vision. Each participant wore a Pico 4 Ultra HMD during the experiment,which provides a 105° fov and a 90 Hz refresh rate. The user-view viewport was rendered at 2048 × 2048 in our experiment. The study was conducted under the supervision and approval of the the Biology and Medical Ethics Committee of Beihang University. All participants provided written informed consent prior to participation.

Conditions. The conditions included our method (SGMRS), Log-Polar (Log-P), and SAT Log-Rectilinear (SLog-R).

Task and Procedure. Using a fixed random seed, we uniformly randomly selected four 360° videos from each of the 4K and 8K datasets. For each selected video, a 10-second segment was extracted by uniformly sampling its start time, resulting in a total of eight video clips. Each participant evaluated the eight clips under four conditions. At the beginning of the experiment for each video clip, participants were first shown a preloaded ground-truth (GT) full-resolution video sequence. They were allowed to freely explore the video in all directions and were informed that this represented the reference quality baseline. Their gaze trajectories were recorded. Subsequently, the video corresponding to the participant's gaze trajectories was generated and transmitted using SGMRS, Log-P, or SLog-R in randomized order. Participants were asked to rate the perceptual visual quality of each sequence. The presentation order of all methods was counterbalanced to ensure balanced exposure across conditions. Perceptual quality was rated on a 5-point Likert scale [5]: 5–no perceptible artifacts, 4–briefly perceptible but acceptable artifacts, 3–noticeable yet acceptable artifacts, 2–clearly noticeable artifacts, 1–severe and disturbing artifacts. To mitigate visual fatigue, participants were given a 30-second rest period after each rating before proceeding to next trial.

Statistical analysis. Subjective quality ratings were analyzed using a two-way repeated-measures ANOVA with Resolution (4K and 8K) and Method (Log-P, SLog-R, and SGMRS) as within-subject factors. For each participant, ratings were averaged across clips within each resolution and method. Significant effects were followed by paired post-hoc comparisons with Holm–Bonferroni correction. Effect sizes were reported using partial eta-squared (η_p^2) for ANOVA effects and Cohen's dz for paired comparisons.

6.2 Results and Discussion

We conducted a two-way repeated-measures ANOVA to analyze the effects of Resolution and Method on subjective quality ratings. The interaction between Resolution and Method was statistically significant ($F_{2,46} = 48.73,\ p < 0.001,\ \eta_p^2 = 0.679$), indicating that the differences among the three methods varied across video resolutions. The main effect of Resolution was also significant ($F_{1,23} = 17.72$, $p < 0.001$, $\eta_p^2 = 0.435$). More importantly, the main effect of Method was significant ($F_{2,46} = 194.80$, $p < 0.001$, $\eta_p^2 = 0.894$), showing that perceived visual quality differed significantly among Log-P, SLog-R, and SGMRS.

Table 5: Subjective quality scores and post-hoc comparisons between SGMRS and the baseline methods.

Res.	Method	$Avg. \pm Std.$	$p_{Holm-corrected}$	Cohen' d_z	Effect Size
4K	Log-P	2.10 ± 0.56	$< 0.001*$	3.18	Huge
	SLog-R	2.96 ± 0.69	$< 0.001*$	3.14	Huge
	SGMRS	4.15 ± 0.50	-	-	-
8K	Log-P	1.85 ± 0.40	$< 0.001*$	3.78	Huge
	SLog-R	2.06 ± 0.72	$< 0.001*$	3.60	Huge
	SGMRS	4.31 ± 0.52	-	-	-

Given the significant Method effect and Resolution-by-Method interaction, we further conducted paired post-hoc comparisons separately for each resolution. As shown in Tab. 5, SGMRS obtained the highest subjective quality scores under both 4K and 8K conditions, and significantly outperformed both Log-P and SLog-R after Holm–Bonferroni correction.

These results demonstrate that SGMRS provides higher perceptual visual quality than the baseline methods. In particular, the stable scores of SGMRS from 4K to 8K suggest that our spherical Gaussian-guided multi-resolution sampling strategy is less dependent on the original video resolution and can deliver stable viewport quality under high-resolution 360° video streaming.

7 Conclusion, Limitations and Future Works

We have presented SGMRS, a spherical Gaussian guided multi-resolution sampling method for foveated 360° video streaming. Unlike prior log-polar and log-rectilinear mappings, which suffer from quality degradation in high-latitude regions due to ERP-induced image distortion, our approach directly models visual acuity on the sphere using a normalized spherical Gaussian distribution. This model guides the construction of a spherical Fibonacci sampling pattern centered on the user's current gaze direction. By precomputing a canonical sampling pattern and exploiting the rotational symmetry of spherical Gaussians, SGMRS enables efficient per-frame sampling and rapid reconstruction under arbitrary gaze motions, while avoiding the need for per-frame re-optimization or complex coordinate transformations. Extensive experiments on 4K and 8K 360° video datasets demonstrate that SGMRS achieves superior perceptual quality in the foveal region while significantly reducing transmission bandwidth, supporting its potential for efficient gaze-adaptive 360° video streaming.

Our method has several limitations. First, although SGMRS targets gaze-adaptive 360° video streaming, the current evaluation does not fully validate a deployed live streaming system. The runtime analysis focuses on the core online sampling and reconstruction stages, without accounting for factors such as live eye-tracking and network transmission delays. Similarly, the user study used prerendered stimuli generated from recorded gaze trajectories rather than a live streaming setup. This design ensured identical gaze paths and viewport contents across methods, improving comparison fairness and controllability, but may not fully reflect live streaming behavior. In addition, the subjective evaluation relied primarily on a single perceived visual quality score and thus did not comprehensively capture factors such as presence, comfort, or overall quality of experience. Future work will evaluate live gaze-contingent streaming systems end to end and incorporate more comprehensive subjective measures. Second, our bandwidth evaluation and coding strategy have limitations. We transmit sampled frames using conventional rectangular video codecs such as H.264. However, the samples follow a sparse, spherical distribution that is not fully suited to standard 2D compression. Although we reorganize the samples into structured frames to improve spatial coherence, H.264 may still introduce encoding overhead. Moreover, evaluating all methods under fixed H.264 settings does not fully reveal their bandwidth efficiency, as different sample organizations may interact differently with the encoder. Future work will explore coding schemes tailored to spherical representations and conduct more comprehensive rate-distortion evaluations. Third, SGMRS currently focuses on single-user foveated 360° video streaming. Future work will extend SGMRS to multi-user settings by constructing a multi-resolution sampling representation that jointly covers multiple gaze directions, enabling shared sampling and reorganization to improve scalability and transmission efficiency.

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REFERENCES

[1] I. Agtzidis, M. Startsev, and M. Dorr. 360-degree video gaze behaviour: A ground-truth data set and a classification algorithm for eye movements. In *Proceedings of the 27th ACM International Conference on Multimedia*, MM '19, p. 1007–1015. Association for Computing Machinery, New York, NY, USA, 2019. doi: 10.1145/3343031.3350947 6

[2] J. Chen, M. Hu, Z. Luo, Z. Wang, and D. Wu. Sr360: Boosting 360-degree video streaming with super-resolution. In *Proceedings of the 30th ACM Workshop on Network and Operating Systems Support for Digital Audio and Video*, pp. 1–6, 2020. 2

[3] M. Dasari, A. Bhattacharya, S. Vargas, P. Sahu, A. Balasubramanian, and S. R. Das. Streaming 360-degree videos using super-resolution. In *IEEE INFOCOM 2020-IEEE Conference on Computer Communications*, pp. 1977–1986. IEEE, 2020. 1, 2

[4] X. Deng, H. Wang, M. Xu, Y. Guo, Y. Song, and L. Yang. Lau-net: Latitude adaptive upscaling network for omnidirectional image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9189–9198, 2021. 1

[5] R. Fan, X. Shi, K. Wang, Q. Ma, and L. Wang. Scene-aware foveated rendering. *IEEE Transactions on Visualization and Computer Graphics*, 30(11):7097–7106, 2024. 9

[6] M. F. Goodchild, P. A. Longley, D. J. Maguire, and D. W. Rhind. Geographic information systems and science. *Wiley & Sons, West Sussex, UK*, 17:517, 2005. 3

[7] Y. Guan, C. Zheng, X. Zhang, Z. Guo, and J. Jiang. Pano: optimizing 360° video streaming with a better understanding of quality perception. In *Proceedings of the ACM Special Interest Group on Data Communication*, SIGCOMM '19, p. 394–407. Association for Computing Machinery, New York, NY, USA, 2019. doi: 10.1145/3341302.3342063 1, 2

[8] B. Guenter, M. Finch, S. Drucker, D. Tan, and J. Snyder. Foveated 3d graphics. *ACM transactions on Graphics (tOG)*, 31(6):1–10, 2012. 2

[9] H. Guo, F. Wang, W. Zhang, Y. Zhu, L. Cui, J. Liu, F. R. Yu, and L. Zhang. Joint adaptation for mobile 360-degree video streaming and enhancement. *IEEE Transactions on Mobile Computing*, 2025. 2

[10] M. Hosseini. View-aware tile-based adaptations in 360 virtual reality video streaming. In *2017 IEEE Virtual Reality (VR)*, pp. 423–424. IEEE, 2017. 1, 2

[11] Z. Jiang, H. Zhu, Y. Lu, G. Ju, and A. Men. Lightweight super-resolution using deep neural learning. *IEEE Transactions on Broadcasting*, 66(4):814–823, 2020. 1, 2

[12] O. Komogortsev and J. Khan. Predictive perceptual compression for real time video communication. In *Proceedings of the 12th Annual ACM International Conference on Multimedia*, pp. 220–227, 2004. 1

[13] C. Ledig, L. Theis, F. Huszár, J. Caballero, A. Cunningham, A. Acosta, A. Aitken, A. Tejani, J. Totz, Z. Wang, et al. Photo-realistic single image super-resolution using a generative adversarial network. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4681–4690, 2017. 2

[14] R. Lee, S. I. Venieris, and N. D. Lane. Deep neural network–based enhancement for image and video streaming systems: A survey and future directions. *ACM Computing Surveys (CSUR)*, 54(8):1–30, 2021. 2

[15] S. Lee, M. S. Pattichis, and A. C. Bovik. Foveated video quality assessment. *IEEE Transactions on Multimedia*, 4(1):129–132, 2002. 7

[16] D. Li, R. Du, A. Babu, C. D. Brumar, and A. Varshney. A log-rectilinear transformation for foveated 360-degree video streaming. *IEEE Transactions on Visualization and Computer Graphics*, 27(5):2638–2647, 2021. 2, 3, 4, 5, 6, 7

[17] B. Lim, S. Son, H. Kim, S. Nah, and K. Mu Lee. Enhanced deep residual networks for single image super-resolution. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pp. 136–144, 2017. 2

[18] H. Liu, W. Ma, Z. Ruan, C. Fang, F. Shang, Y. Liu, L. Wang, C. Wang, and D. Jiang. A single frame and multi-frame joint network for 360-degree panorama video super-resolution. *Engineering Applications of Artificial Intelligence*, 134:108601, 2024. 1

[19] X. Liu, Y. Huang, L. Song, R. Xie, and X. Yang. The sjtu uhd 360-degree immersive video sequence dataset. In *2017 International Conference on Virtual Reality and Visualization (ICVRV)*, pp. 400–401, 2017. doi: 10.1109/ICVRV.2017.00095 6

[20] P. Lungaro, R. Sjöberg, A. J. F. Valero, A. Mittal, and K. Tollmar. Gaze-aware streaming solutions for the next generation of mobile vr experiences. *IEEE transactions on visualization and computer graphics*, 24(4):1535–1544, 2018. 1, 2

[21] Z. Luo, B. Chai, Z. Wang, M. Hu, and D. Wu. Masked360: Enabling robust 360-degree video streaming with ultra low bandwidth consumption. *IEEE Transactions on Visualization and Computer Graphics*, 29(5):2690–2699, 2023. 2

[22] R. K. Mantiuk, G. Denes, A. Chapiro, A. Kaplanyan, G. Rufo, R. Bachy, T. Lian, and A. Patney. Fovvideovdp: A visible difference predictor for wide field-of-view video. *ACM Transactions on Graphics (TOG)*, 40(4):1–19, 2021. 7

[23] X. Meng, R. Du, M. Zwicker, and A. Varshney. Kernel foveated rendering. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 1(1):1–20, 2018. 2, 3, 6

[24] A. T. Nasrabadi, A. Mahzari, J. D. Beshay, and R. Prakash. Adaptive 360-degree video streaming using layered video coding. In *2017 IEEE Virtual Reality (VR)*, pp. 347–348. IEEE, 2017. 1, 2

[25] D. V. Nguyen, H. T. Tran, and T. C. Thang. A client-based adaptation framework for 360-degree video streaming. *Journal of Visual Communication and Image Representation*, 59:231–243, 2019. 1

[26] A. Nishiyama, S. Ikehata, and K. Aizawa. 360 single image super resolution via distortion-aware network and distorted perspective images. In *2021 IEEE International Conference on Image Processing (ICIP)*, pp. 1829–1833. IEEE, 2021. 1

[27] C. Ozcinar, A. De Abreu, and A. Smolic. Viewport-aware adaptive 360 video streaming using tiles for virtual reality. In *2017 IEEE International Conference on Image Processing (ICIP)*, pp. 2174–2178. IEEE, 2017. 2

[28] J. Park and K. Nahrstedt. Navigation graph for tiled media streaming. In *Proceedings of the 27th ACM International Conference on Multimedia*, pp. 447–455, 2019. 2

[29] F. Qian, B. Han, Q. Xiao, and V. Gopalakrishnan. Flare: Practical viewport-adaptive 360-degree video streaming for mobile devices. In *Proceedings of the 24th Annual International Conference on Mobile Computing and Networking*, pp. 99–114, 2018. 1, 2

[30] F. Qian, L. Ji, B. Han, and V. Gopalakrishnan. Optimizing 360 video delivery over cellular networks. In *Proceedings of the 5th Workshop on All Things Cellular: Operations, Applications and Challenges*, pp. 1–6, 2016. 1

[31] S. Rimac-Drlje, M. Vranješ, and D. Žagar. Foveated mean squared error—a novel video quality metric. *Multimedia tools and applications*, 49(3):425–445, 2010. 7

[32] J. Ryoo, K. Yun, D. Samaras, S. R. Das, and G. Zelinsky. Design and evaluation of a foveated video streaming service for commodity client devices. In *Proceedings of the 7th International Conference on Multimedia Systems*, pp. 1–11, 2016. 1

[33] M. Saraiji, K. Minamizawa, and S. Tachi. Foveated streaming: Optimizing video streaming for telexistence systems using eye-gaze based foveation. *Virtual Reality Society of Japan*, 1(2), 2017. 2

[34] Z. Shen, C. Lin, K. Liao, L. Nie, Z. Zheng, and Y. Zhao. Panoformer: panorama transformer for indoor 360° depth estimation. In *European Conference on Computer Vision*, pp. 195–211. Springer, 2022. 1

[35] Z. Shen, C. Lin, L. Nie, K. Liao, and Y. Zhao. Neural contourlet network for monocular 360 depth estimation. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(12):8574–8585, 2022. 1

[36] H. Strasburger, I. Rentschler, and M. Jüttner. Peripheral vision and pattern recognition: A review. *Journal of vision*, 11(5):13–13, 2011. 4

[37] S. Winkler, E. D. Gelasca, and T. Ebrahimi. Toward perceptual metrics for video watermark evaluation. In *Applications of Digital Image Processing XXVI*, vol. 5203, pp. 371–378. SPIE, 2003. 7

[38] E. S. Wong, N. H. A. Wahab, F. Saeed, and N. Alharbi. 360-degree video bandwidth reduction: Technique and approaches comprehensive review. *Applied Sciences*, 12(15):7581, 2022. 2

[39] C. Wu, Z. Wang, and L. Sun. Paas: A preference-aware deep rein-

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forcement learning approach for 360 video streaming. In *Proceedings of the 31st ACM Workshop on Network and Operating Systems Support for Digital Audio and Video*, pp. 34–41, 2021. 2

[40] A. Yaqoob and G.-M. Muntean. A combined field-of-view prediction-assisted viewport adaptive delivery scheme for 360° videos. *IEEE Transactions on Broadcasting*, 67(3):746–760, 2021. 2

[41] J. Ye, A. Xie, S. Jabbireddy, Y. Li, X. Yang, and X. Meng. Rectangular mapping-based foveated rendering. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 756–764. IEEE, 2022. 2

[42] H. Yeo, C. J. Chong, Y. Jung, J. Ye, and D. Han. Nemo: enabling neural-enhanced video streaming on commodity mobile devices. In *Proceedings of the 26th Annual International Conference on Mobile Computing and Networking*, pp. 1–14, 2020. 2

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